MSc Data Science Dissertation

Final Group Project Presentation

# Edge-Deployable Rock Detection & Burial-Status Classification

## Real-Time Embedded Computer Vision for Precision Agriculture on Raspberry Pi 5

**Project Team: 1. Tanishk Nanasaheb Shinde 2. Megh Kashilkar 3. Sanskar Sanju Gade 4. Jitendra Suwalka**

**Academic Supervisor:** Dr. Felipe Campelo (University of Bristol)

**External Supervisors:** Dr. Ahmad Al-Mallahi & Jason McLaren (Dalhousie University / McCain Foods)

University of Bristol | Faculty of Engineering | Department of Engineering Mathematics | 16 September 2026

# Rocks Cause Severe Operational & Equipment Costs in Potato Farming

University of BRISTOL

## Operational Challenge & Need

- Field rocks obstruct tillage, damage mechanical harvester blades, and puncture sorting equipment.
- Harvesting downtime and manual rock picking represent significant commercial bottlenecks.
- Removing rocks requires knowing both **location** and **burial status** (completely exposed vs half-buried) to calibrate actuation force.

## Industry Collaboration Objectives

- **Partner:** McCain Foods & Dalhousie University.
- **Goal:** Real-time vision deployed directly on field tractors.
- **Constraint:** Rural fields lack continuous cloud connectivity, demanding **fully offline edge inference**.
- **Target:** Low-cost embedded compute (Raspberry Pi 5).

# High Model Accuracy Does Not Guarantee Real-Time Edge Feasibility

## The Academic vs Edge Deployment Gap

- Standard research relies on GPU servers with FP32 precision.
- Tractor operations at 5–10 km/h require $\geq$ **15–20 FPS** for real-time hydraulic rock picker guidance.
- Baseline PyTorch on Raspberry Pi 5 runs at only **2.52 FPS (397 ms latency)** — unusable for live control.
- Quantisation (FP16/INT8) and embedded runtimes (TFLite/NCNN) are mandatory.

## Edge Physical Constraints

- **Compute:** Quad-core ARM Cortex-A76 without discrete GPU.
- **Power & Heat:** Enclosed tractor cabs with passive cooling risk thermal CPU throttling.
- **Memory:** Must operate reliably within limited RAM without paging.
- **Stability:** 100% crash-free continuous execution required.

# Core Research Question & Empirical Hypotheses

University of BRISTOL

## Central Research Question

*"How effectively can lightweight YOLO models be optimised, quantised, and deployed for real-time rock detection and burial-status classification on a low-cost edge device (Raspberry Pi 5)?"*

- **Q1:** What is the best architecture for exposed vs half-buried rocks?
- **Q2:** What are the empirical trade-offs between precision, speed, memory, and stability?

## Guiding Hypotheses

- **H1 (Speedup):** Quantised runtimes (TFLite INT8 / NCNN) will yield $\geq 5\times$ latency reduction over PyTorch baseline.
- **H2 (Accuracy Cost):** INT8 quantisation will incur a measurable localisation penalty ($\text{mAP}drop < 5\%$).
- **H3 (Operational Divergence):** Runtimes will show significant divergence in CPU load, thermals, and endurance stability.

# Rigorous Five-Stage End-to-End Project Architecture

## 5-Stage Execution Pipeline

1. **Leakage-Free Splitting:** 313 Train / 45 Val / 90 Test images partitioned *before* augmentation.
2. **Controlled Augmentation:** 6 safe photometric transforms expanding training to 2,191 images (3,563 rocks).
3. **Controlled Detector Selection:** YOLOv8n, YOLO11n, YOLO11s trained identically; selection via validation mAP50–95.
4. **Six-Format Edge Export:** PyTorch, TorchScript, ONNX, TFLite (FP32/INT8), NCNN (FP32/FP16).
5. **Pi 5 Benchmark:** Fixed 90-image test + 10-pass endurance run measuring speed, quality, RAM, and thermals.

## Strict Experimental Controls

- **Zero Test Leakage:** Test partition remained untouched throughout development.
- **Constant Neural Weights:** Frozen weights exported across all six runtimes.
- **Standard Letterboxing:** $640 \times 640$ with symmetric fill (value 114).
- **Systematic Telemetry:** CPU, RAM, and thermals tracked at 10 ms intervals.

# Original Field Dataset: 448 Images & 706 Annotated Rocks

## Class Balance & Distribution

- **448 field images** collected from real potato cultivation plots.
- **706 ground-truth rock annotations**:
  - **Completely Exposed:** 362 objects (51.3%) — clear boundaries, high surface contrast.
  - **Half-Buried:** 344 objects (48.7%) — partial soil occlusion, irregular silhouettes.
- Closely balanced (51.3% vs 48.7%) totals allow performance differences to be interpreted as visual difficulty rather than class imbalance.

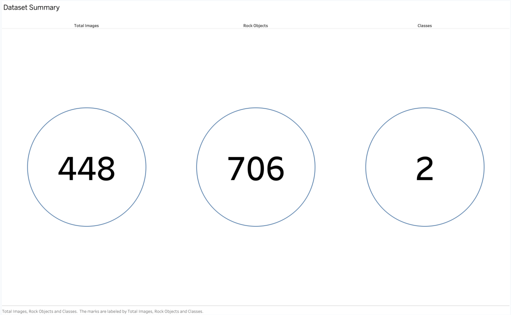


Figure 3.4: Dataset summary showing 448 images and 706 annotated rocks.

# Pre-Augmentation Splitting Prevents Data Leakage

## Partitioning Strategy

- Split performed **prior to augmentation** to ensure zero augmented copies leaked into validation or test sets.
- **Training (313 images):** 259 exposed, 250 half-buried rocks (69.9%).
- **Validation (45 images):** 34 exposed, 28 half-buried rocks (10.0%).
- **Test (90 images):** 69 exposed, 66 half-buried rocks (20.1%).
- Preserves absolute independence for edge benchmarking.

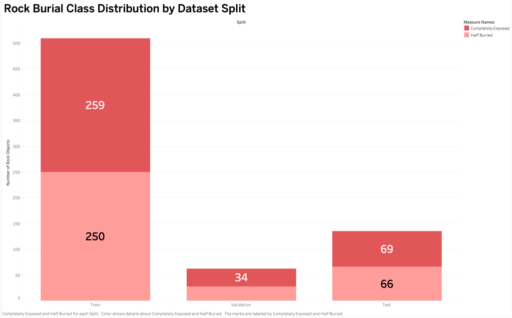


Figure 3.2: Class-level annotation counts across Train, Val, and Test splits.

# Controlled Photometric Augmentation Expands Training to 2,191 Images

## Safe Photometric Transformation

- Each of the 313 training images received 6 photometric variants (+1,878 images), yielding **2,191 training images**.
- Training annotations expanded from 509 to **3,563 rock instances**.
- **Transforms:** Brightness/contrast shifts, gamma adjustments, and HSV colour jitter to simulate lighting changes.
- **Geometric Safety:** Extreme rotations/crops strictly avoided to maintain bounding box accuracy.

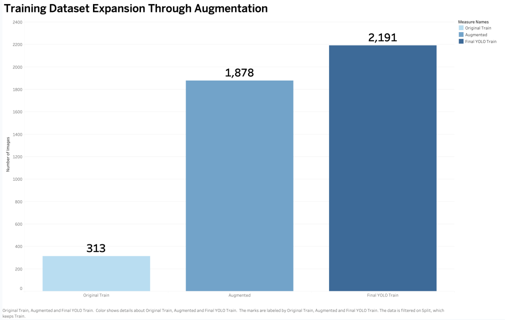


Figure 3.3: 7x expansion of the training partition from 313 to 2,191 images.

# Single-Stage YOLO Detects Location & Burial Status Simultaneously

## Why Single-Stage YOLO for Edge Vision?

- Two-stage detectors (Faster R-CNN) require separate region proposal networks ($>$ 800 ms on ARM CPU).
- Single-stage YOLO predicts bounding boxes $[x, y, w, h]$ and class probabilities in a **single forward pass**.
- Modern anchor-free architecture adapts to irregular rock shapes and small objects without manual anchor tuning.

## Output Processing Pipeline

- **Input:** $640 \times 640 \times 3$ RGB tensor.
- **Decoded Output:** 8,400 candidate anchor predictions.
- **Confidence Threshold:** Filtering scores $< 0.25$.
- **NMS:** IoU threshold 0.70 removes duplicate detections.
- **Telemetry Output:** Box coordinates, class labels, inference latency, and aggregate rock counts per image.

# Controlled Training of Lightweight YOLO Candidates

## Experimental Setup

- Evaluated three architectures: **YOLOv8n** (3.2M params), **YOLO11n** (2.6M params), and **YOLO11s** (9.4M params).
- **Identical Hyperparameters:** 20 epochs, AdamW optimiser, Cosine LR scheduler, batch size 32, AMP enabled.
- Pre-trained COCO transfer learning initialisation.
- Validation evaluation at 1024-pixel resolution.

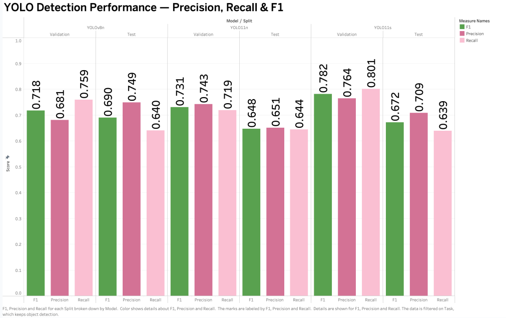


Figure 3.5: Precision, Recall, and F1 across candidate YOLO architectures.

# Predefined Primary Criterion: Validation mAP50–95

## Rigorous Selection Standard

- Architecture selection governed strictly by **Validation mAP50–95**.
- **Why mAP50–95 over mAP50?**
  - mAP50 evaluates loose overlap (IoU $\geq$ 0.50).
  - mAP50–95 averages across IoU 0.50 to 0.95, strictly penalising poor bounding box localisation.
- Accurate boundary estimation is vital for mechanical rock pickers to avoid striking stone edges.

## Candidate Validation Scores

| Model | F1 | mAP50 | mAP50–95 |
|---|---|---|---|
| **YOLOv8n** | 0.718 | 0.759 | **0.4288** |
| YOLO11n | 0.731 | 0.729 | 0.4002 |
| YOLO11s | **0.782** | **0.783** | 0.4107 |

**Decision:** YOLOv8n selected based on the predefined primary criterion and compact 3.2M parameter size.

# YOLOv8n Achieves Highest Validation Localisation Quality

## Selection Summary

- YOLOv8n achieved **mAP50–95 = 0.4288**, outperforming YOLO11n (0.4002, +7.1%) and YOLO11s (0.4107, +4.4%).
- Although YOLO11s scored slightly higher on F1 (0.782), its 9.4M parameter size tripled compute requirements.
- YOLOv8n provided the optimal balance of localisation precision and lightweight footprint.
- Checkpoint weights frozen for multi-format edge export.

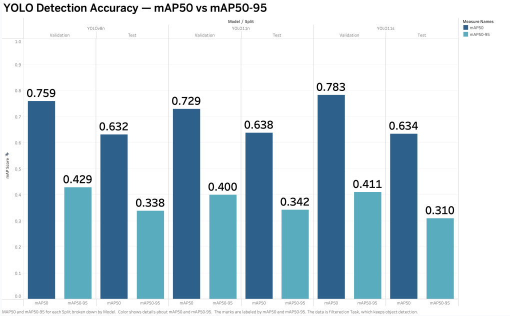


Figure 3.6: mAP50 and mAP50–95 on validation and test partitions.

# Class Breakdown Reveals Inherent Difficulty of Half-Buried Rocks

## Class-Specific Asymmetry on Test Set

- **Completely Exposed:** 64 / 69 rocks detected correctly (**92.8% operational accuracy**).
- **Half-Buried:** Only 23 / 66 rocks detected correctly (**34.8% operational accuracy**).
- **Failure Causes:**
  - Gradual surface-to-soil boundary blending.
  - Partial soil occlusion and shadow contours.
  - Absence of continuous high-contrast edges.

## Operational Implication

- Exposed rocks are detected with $>90\%$ reliability for fast surface sweeps.
- Half-buried rocks require deeper penetration and higher actuation force; classification errors risk machinery jams.
- 2D RGB imagery alone reaches a performance ceiling; future 3D sensing is recommended.

# CLIP Enhances Crop Classification but Lacks Native Localisation

University of BRISTOL

## Zero-Shot vs Fine-Tuned CLIP (ViT-B/32)

- Investigated whether foundation vision-language models could improve burial classification on cropped rock chips.
- **Zero-Shot CLIP:** 50.4% accuracy (chance level between exposed vs buried).
- **Fine-Tuned CLIP: 80.0% accuracy** on cropped rock chips.
- Demonstrates strong semantic feature representations once fine-tuned on agricultural data.

## Why CLIP Was Not Deployed to Edge

- CLIP is a crop classifier, **not an object detector**; requires a separate detector to provide bounding boxes.
- Two-stage pipeline doubles complexity.
- ViT-B/32 requires $\sim$ 350 MB RAM and $\sim$ 850 ms latency per crop on Raspberry Pi 5 — far too slow for real-time control.

# End-to-End Edge Pipeline: Letterboxing to Count Telemetry

## Data Flow on Raspberry Pi

1. **Letterboxing:** $640 \times 640$ with aspect-ratio preservation and fill value 114.
2. **Tensor Conversion:** RGB normalization $[0, 1]$ and NCHW formatting.
3. **Engine Inference:** Multi-backend forward pass (PyTorch, TFLite, NCNN, ONNX).
4. **Post-Processing:** Threshold filtering ($\geq 0.25$) and NMS (IoU 0.70).
5. **Output Telemetry:** Bounding boxes, labels, FPS, and rock count MAE.

## Rock Counting Mean Absolute Error (MAE)

- **PyTorch FP32:** MAE $= 0.600$ rocks/image
- **TFLite FP32:** MAE $= 0.600$ rocks/image
- **TFLite INT8:** MAE $= 0.633$ rocks/image ($+0.033$ error delta)
- **NCNN FP32:** MAE $= 0.600$ rocks/image
- **NCNN FP16:** MAE $= 0.611$ rocks/image

INT8 quantisation retains excellent rock-counting fidelity across test images.

# Qualitative Failure Modes: Soil Clods & Occlusion Boundaries

## Systematic Error Patterns

- **False Positives (Soil Clods):** Dry, crusted soil clumps frequently exhibit convex contours and shadow gradients mimicking small rocks.
- **False Negatives (Deep Burial):** Rocks with $< 20\%$ surface exposure blend into tillage patterns.
- **Boundary Oscillation:** Stones with $\sim 50\%$ burial fluctuate between exposed and half-buried classes.

## Diagnostic Insights

- **Illumination Robustness:** Photometric augmentation successfully prevented false alarms under harsh direct sunlight.
- **Scale Sensitivity:** Rocks $<$ 30 pixels across exhibit lower confidence; $640 \times 640$ represents the optimal speed-resolution trade-off.
- Suggests integrating 3D surface depth profiling in future tractor hardware revisions.

# Preparation of Frozen Checkpoint & Calibration Set

## Checkpoint & Calibration Design

- Master YOLOv8n weights frozen into `best.pt` (6.2 MB).
- **INT8 Calibration Dataset:**
  - Extracted 200 original training images containing 304 rock objects.
  - Balanced at **152 exposed and 152 half-buried** instances.
  - Computed static scale factor $S$ and zero-point $Z$ without backpropagation.
- Zero test data observed during calibration.

## Affine Quantisation Formula

$$q = \text{round}\left(\frac{r}{S}\right) + Z$$

- $r$: 32-bit floating point weight/activation.
- $q$: 8-bit signed integer $[-128, 127]$.
- $S$: positive real scale factor; $Z$: integer zero point.
- Unlocks ARM NEON integer dot-product SIMD instructions for massive edge speedups.

# Six Deployment Representations Span FP32, FP16 & INT8

## Evaluated Edge Representations

- **PyTorch FP32 (6.2 MB):** Native baseline runtime.
- **TorchScript FP32 (12.3 MB):** JIT-compiled PyTorch IR.
- **ONNX FP32 (12.2 MB):** Open Neural Network Exchange.
- **TFLite FP32 (12.0 MB):** TensorFlow Lite float engine.
- **TFLite INT8 (3.3 MB):** 8-bit quantized flatbuffer (**–73% file size**).
- **NCNN FP32 (12.1 MB) & FP16 (6.1 MB):** High-performance mobile ARM engine.

## Multi-Format Scope

- All six representations exported from identical frozen weights.
- File sizes span from 12.3 MB down to 3.3 MB.
- Enables rigorous cross-comparison across runtime engines (PyTorch vs TFLite vs NCNN vs ONNX) and numerical precisions (FP32 vs FP16 vs INT8).

# Hardware Testbed: Raspberry Pi 5 (8GB) in CPU-Only Mode

## Raspberry Pi 5 Specifications

- **SoC:** Broadcom BCM2712 quad-core ARM Cortex-A76 @ 2.4 GHz.
- **Memory:** 8 GB LPDDR4X SDRAM.
- **Power:** Official 27W (5.1V / 5.0A) USB-C power supply.
- **OS:** Debian GNU/Linux 13 (Trixie), 64-bit kernel.
- **Cooling:** Passive heatsink only (No Active Cooler) to simulate enclosed tractor equipment boxes.
- **Stack:** Python 3.13.5, Ultralytics 8.4.115, ONNX Runtime 1.28.0, OpenCV 5.0.0.

## Experimental Protocol Controls

- Headless OS environment without desktop GUI jitter.
- Batch size = 1 strictly enforced (live video simulation).
- Hardware telemetry: psutil and vcgencmd polled at 10 ms intervals for CPU load, RSS memory, and SoC core temperature.

# Dual Testing Protocol: Fixed Benchmark & Continuous Endurance

## Phase 1: Canonical Fixed Benchmark

- 3 warmup inferences to stabilize cache and JIT execution.
- 90 timed sequential inferences over held-out test set.
- Measures per-frame latency, median FPS, CPU %, peak RAM, mAP50, mAP50–95, and Count MAE.

## Phase 2: Continuous Endurance Test

- 900 consecutive inferences (10 passes of the 90-image test set).
- Tests memory leak accumulation, thermal throttling, and runtime stability under prolonged execution.
- **Insight:** Models can pass a 90-image benchmark but crash after 20 minutes in field conditions.

# TFLite INT8 Delivers 22.62 FPS — 9.01x Speedup over PyTorch

## Canonical Runtime Latency & Throughput

- **PyTorch FP32:** 397.48 ms | 2.52 FPS (Baseline)
- **TFLite FP32:** 143.70 ms | 6.95 FPS (2.77× speedup)
- **NCNN FP32:** 75.09 ms | 13.26 FPS (5.29× speedup)
- **NCNN FP16:** 72.31 ms | 13.82 FPS (5.49× speedup)
- **TFLite INT8 (WINNER): 44.13 ms | 22.62 FPS** (9.01× **speedup!)**

TFLite INT8 is the **only** representation that exceeds the 20 FPS threshold required for real-time field operations.

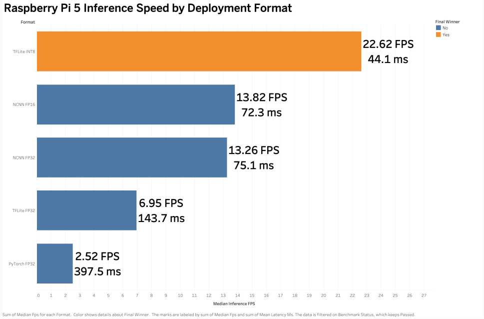


Figure 4.3: Median throughput (FPS) and mean latency (ms) across formats on Pi 5.

# Throughput vs Accuracy: A Quantifiable, Acceptable Trade-Off

## Speed vs Accuracy Trade-off

- **PyTorch FP32:** mAP50-95 = 0.275, F1 = 0.622, FPS = 2.52
- **TFLite FP32:** mAP50-95 = 0.270, F1 = 0.599, FPS = 6.95
- **NCNN FP16:** mAP50-95 = 0.270, F1 = 0.600, FPS = 13.82
- **TFLite INT8:** mAP50-95 = 0.234, F1 = 0.594, **FPS = 22.62**
- **Trade-off Assessment:** A 14.9% drop in mAP50–95 buys an **800% gain in throughput** (2.5 → 22.6 FPS). Highly favorable for field deployment.

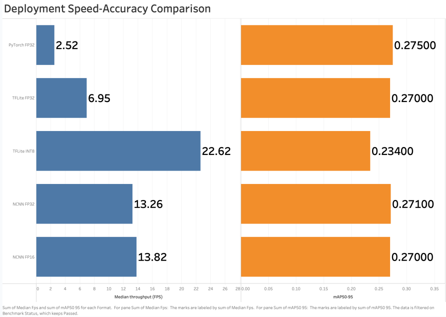


Figure 4.2: Median throughput vs mAP50–95 Pareto scatter plot.

# System Resources: INT8 Optimises CPU, Memory & Thermal Load

## Hardware Utilisation on Raspberry Pi 5

- **Peak Process Memory:** TFLite INT8 used the least RAM: **824.2 MB** (vs 868.3 MB for TFLite FP32).
- **CPU Core Load:** TFLite INT8 required **62.8%** CPU load, leaving headroom for camera capture and telemetry.
- **Thermal Dissipation:** TFLite INT8 ran coolest at **55.10 °C** (vs 66.10 °C for TFLite FP32).
- Integer SIMD instructions generate significantly less thermal dissipation.

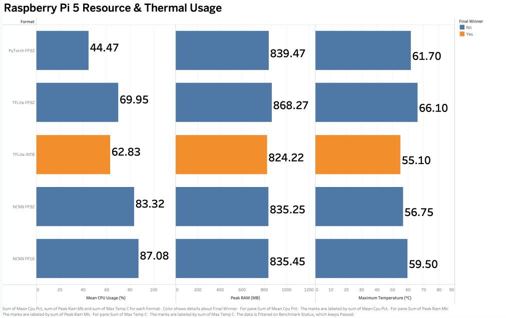


Figure 4.5: CPU load, peak RAM, and short-run temperature across formats.

# Stability Testing Exposes Critical Runtime Failures

University of BRISTOL

## Negative Results & Runtime Crashes

- **Export success does not guarantee edge stability!**
- **ONNX FP32 Fatal Failure:** Exported cleanly, but crashed completely during benchmark aggregation on ARM64.
- **NCNN Endurance Crash:** Passed 90-image benchmark, but crashed during continuous 10-pass endurance run due to thread contention.

## Robustness & Deployment Findings

- **TFLite INT8 Winner:** The **only** format that completed 100% of both fixed benchmark and continuous endurance runs without failure.
- **System-Level Validation:** Highlights the critical necessity of measuring on target physical hardware under real workload stress.
- **Reporting Integrity:** Disclosing runtime failures prevents flawed assumptions about portable model equivalence.

# Final Selection: TFLite INT8 Validated; Future Roadmap Defined

## Final Selection: TFLite INT8

- **Speed:** 22.62 FPS ($9.01\times$ speedup, $> 20$ FPS threshold).
- **Localisation:** Retains 0.234 mAP50–95, 0.633 count MAE.
- **Efficiency:** Lowest RAM (824 MB) and lowest temperature (55.1 °C).
- **Reliability:** 100% stability across all continuous runs.
- **Hypotheses Supported:** H1 (Speedup), H2 (Accuracy Cost), and H3 (Stability Divergence) fully validated.

## Prioritised Future Roadmap

1. **3D Depth Sensing:** Integrate stereo vision / LiDAR to elevate half-buried detection above 35%.
2. **Live Video Stream:** Build GStreamer pipeline with GPS geo-tagging.
3. **NPU Acceleration:** Deploy on Raspberry Pi AI Kit (Hailo-8L NPU, 13 TOPS) for $< 10$ ms latency.
4. **Actuator Integration:** Connect to hydraulic rock picker PLC via CAN bus.